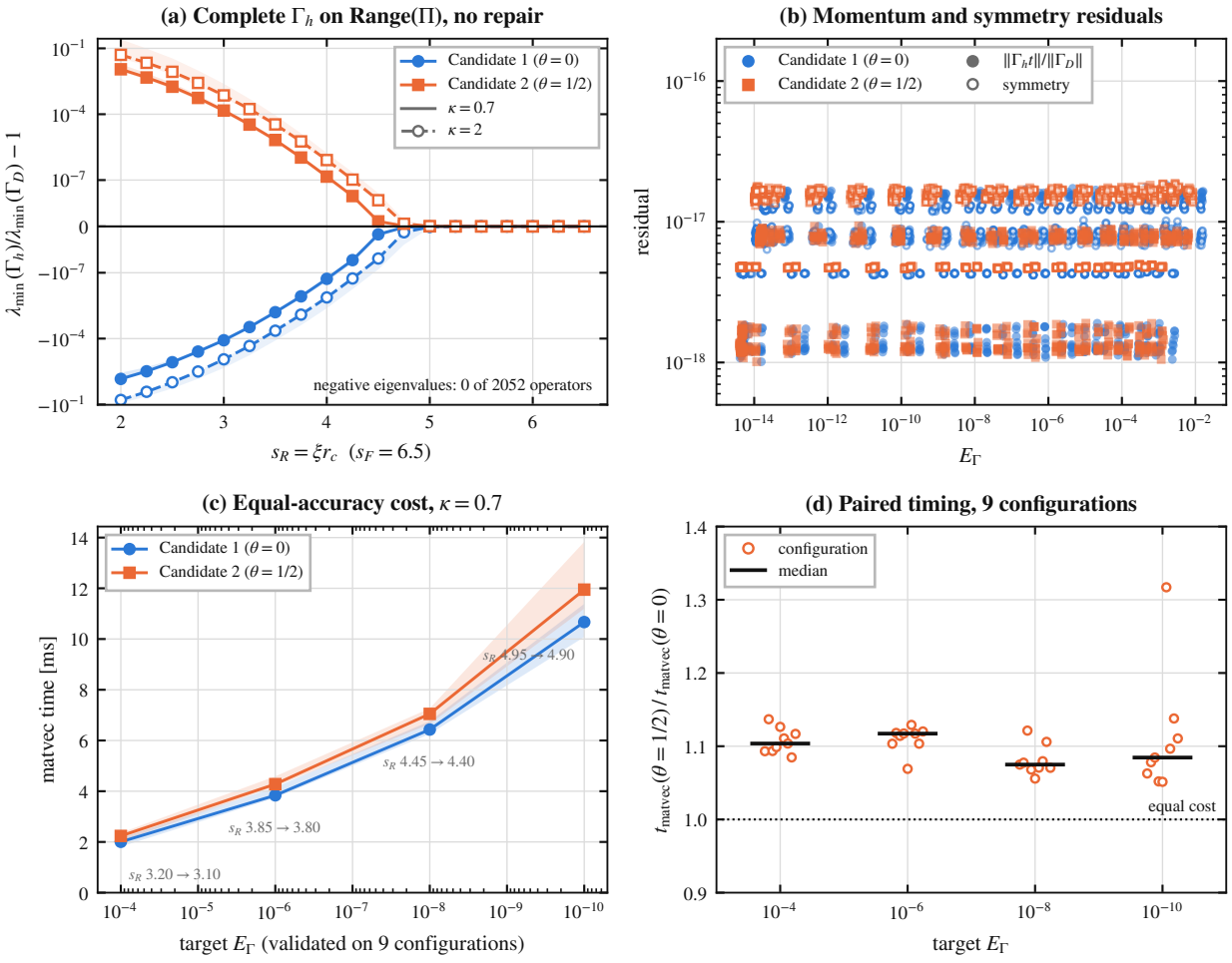

(a) Complete Γ_h on Range(Π), no repair
Candidate 1 ($\theta = 0$)
Candidate 2 ($\theta = 1/2$)
$\kappa = 0.7$
$\kappa = 2$
negative eigenvalues: 0 of 2052 operators
$\lambda_{\min}(\Gamma_h)/\lambda_{\min}(\Gamma_D) - 1$
$s_R = \xi r_c$ ($s_F = 6.5$)
(b) Momentum and symmetry residuals
Candidate 1 ($\theta = 0$)
Candidate 2 ($\theta = 1/2$)
$\|\Gamma_h l\|/\|\Gamma_D\|$
symmetry
residual
E_Γ
(c) Equal-accuracy cost, $\kappa = 0.7$
Candidate 1 ($\theta = 0$)
Candidate 2 ($\theta = 1/2$)
s_R 4.95 → 4.90
s_R 4.45 → 4.40
s_R 3.85 → 3.80
s_R 3.20 → 3.10
matvec time [ms]
target E_Γ (validated on 9 configurations)
(d) Paired timing, 9 configurations
configuration
median
equal cost
$t_{\text{matvec}}(\theta = 1/2)\,/\,t_{\text{matvec}}(\theta = 0)$
target E_Γ